AI Used for Deliverable 1

Kiki

Here's a summary of what we covered, all around pandas and matplotlib plotting.

1. `DataFrame.append` **error:** It was removed in pandas 2.0. Use `pd.concat([df, pd.DataFrame([row])], ignore_index=True)`, and in loops collect rows in a list first and concat once.

2. Combined bar and line chart with a legend: Bars and line sit on different axes (`twinx`), so you gather handles from both with `get_legend_handles_labels()` and build one legend on `ax1`. Red bars were labeled volume, blue line precip.

3. Legend outside the plot: Use `loc` with `bbox_to_anchor`, e.g. `bbox_to_anchor=(0.5, -0.15)` for below the plot, or `(1.1, 1)` for the right side. `plt.tight_layout()` or `savefig(..., bbox_inches='tight')` prevents cutoff.

4. Getting the year from a datetime: `.dt.year` on a pandas column. Your format (`2025-01-02T00:00:00.000Z`) parses directly with `pd.to_datetime`, optionally with `utc=True`.

5. Removing the year from the x-axis: Options were `strftime('%m-%d')` for a string column, `DateFormatter('%m-%d')` for display only, or a dummy year (2000) to overlay multiple years.

6. Overlapping x ticks: Rotate with `fig.autofmt_xdate()`, or use locators like `MonthLocator()` with `DateFormatter('%b')` and `set_xlim` to stop at Dec 31.

7. Your final constraint: `month_day` is an object (string) column and you don't want to add a dummy year. Since string axes get one tick per value, the fix is to thin the ticks manually:

- every Nth tick: `ax.set_xticks(ax.get_xticks()[::15])`
- one tick per month, using the first entry of each month (positions where the month part changes)
- or `plt.MaxNLocator(12)` to cap the number of ticks

Thea

1. Limiting values on x-axis:

```
# Rotate the x-axis values and limit to 24
plt.xticks(rotation=45)
ax.xaxis.set_major_locator(ticker.MaxNLocator(nbins=24))
```

2. Explaining pandas rolling function to use in task 5